

Chapter: 1

Definition of Statistics:-

[Statistics is that branch of knowledge that deal with collection, organization, presentation, analysis and interpretation of the numerical data.]

According to Dr. AL Bowley, Statistics are numerical statement of fact in any department of enquiry placed in relation to each other.

Importance in CSE:-

It is very important for a student of computer sciences. As computer sciences also deal with organization and interpretation of numerical facts. In fact most of the principles of computer sciences are based on concepts of statistics.

The computer can process large amount of data quickly and accurately. For the large number of data some of important statistical packages that have been used are - SPSS, SAS, STATA, S-plus and MINITAB.

Q1 Definition of variable with example:-

Any phenomenon which varies from individual to individual is called variable. Variable are represented by symbols (e.g. x, y or z) for example Age, weight, height, sex, income and expenses, country of birth, eye colour, gender are example of variables.

Q1 Variable are two types:-

1. Qualitative.

2. Quantitative.

Qualitative Variable:-

A variable which can not expressed as numerically is called qualitative (categorical) variable.

Example :- Hair Color, gender, Programming Language, brand etc.

Quantitative Variable:-

A variable which can expressed as numerically.

Examp $\Rightarrow$ Quantitative Variable are Two types.

i) Discrete

ii) Continuous.

Discrete Variable :-

A variable is called discrete variable if it can take only isolated (whole) values.

Example :- Family member, number of cars etc.

Continuous variable :-

A variable is called continuous variables if it can take any values between certain limits.

Example :- Height, weight, temperature.

Q1 Discrete and Continuous Variable :-

Discrete	Continuous
A variable is called discrete variable if it can take only isolated values	A variable is called continuous variable if it can take any values between certain limits.

It is countable	It is measurable
The values of discrete variable may be finite or infinite	It must be infinite
Family member, student member	age, weight, height, salary etc.

$$P_{78} = 30 + \frac{\frac{78 \times 30}{100} - 23}{5}$$

$$= 32.88$$

Chapter 1

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- i) The temperature of CPU or server {continuous}
- ii) The amount of memory being used by a process {discrete}
- iii) Program execution time {continuous}
- iv) The count of clicks on a JIUC webpage {discrete}.
- v) The programming language being used in a project (qualitative).

Chapter 2

3 Geometric Mean:-

For Group Data,

$$G.M = \left(x_1^{f_1} \cdot x_2^{f_2} \cdots x_n^{f_n} \right)^{\frac{1}{\sum f_i}}$$

For Ungroup Data,

$$G.M = \left(x_1 \cdot x_2 \cdots x_n \right)^{\frac{1}{n}}$$

Let If Given data, = {2, 1, 6}

$$\text{Then, } G.M = (2 \times 1 \times 6)^{\frac{1}{3}}$$

$$= (12)^{\frac{1}{3}} = \sqrt[3]{12}$$

$$= 2.289.$$

Median:-

For Group data,

$\frac{n}{2}$

$$\text{Mean} = L +$$

$$\frac{\frac{n}{2} - F_c}{f_m} \times c$$

$$\text{Median} = L +$$

Let if Given data,

	f_c	$F.c$
0-5	2	2
5-10	4	6
10-15	3	9
15-20	8	17
20-25	1	18

$$\text{Median} = 10 + \frac{9-6}{3} \times 5$$
$$= 15$$

For Ungrouped data,

If number of data is odd,

$$\text{Median} = \frac{n+1}{2} \text{th observation.}$$

If the given data = $\{6, 5, 3, 7, 1\}$

For First we arrange this data,

$$\text{So, } = \{1, 3, 5, 6, 7\}$$

$$\text{Median} = \frac{5+1}{2} \text{th observation.}$$

$$= 3$$

So, Median is 5.

If the number of data is even,

$$\text{Median} = \frac{\frac{n}{2} + \left(\frac{n+1}{2}\right)}{2} \text{th observation.}$$

If the given data: 15, 10, 13, 5

Firstly we arrange the observation by ascending order of magnitude:-

5, 10, 13, 15

$$\text{Median} = \frac{\frac{4}{2} + \left(\frac{4+1}{2}\right) \left(\frac{4}{2} + 1\right)}{2}$$

$$= \frac{2\text{th observation} + 3\text{th observation}}{2}$$

$$= \frac{10 + 13}{2}$$

$$= 11.5$$

Mode :-

For Group data,

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$$

Let if Given data,

Number of Frames.	f_c
150 - 155	5
155 - 160	2
160 - 165	7
165 - 170	10

$$\text{Mode} = 165 + \frac{3}{3+10} \times 5$$

$$= 166.153$$

For Ungroup data,

Which frequency is maximum.

Let if Given data,

2, 5, 9, 5, 3, 5.

Here maximum frequency is of observation of 5. So mode is 5.

Formulae for computing,

$$\text{Quantile, } Q_i = L + \frac{\frac{i \times n}{4} - F_o}{f_o} \times c$$

$$\text{Decile, } D_i = L + \frac{\frac{i \times n}{10} - F_o}{f_o} \times c$$

$$\text{Percentiles, } P_i = L + \frac{\frac{i \times n}{100} - F_p}{f_p} \times c$$

(4)

R.T (ms)	$F_2 (f_i)$	F.C	x_i	$f_i x_i$
0-10	3	3	5	15
10-20	8	11	15	120
20-30	12	23	25	300
30-40	5	28	35	175
40-50	2	30	45	90
		$n = 30$		$\Sigma f_i x_i = 700$

(i)

~~$$\text{Mean} = L + \frac{A_1}{A_1 + A_2} \times c$$~~

$$\text{Mean} = \frac{\Sigma f_i x_i}{n} = \frac{700}{30} = 23.33$$

$$\Rightarrow \text{Median} = L + \frac{\frac{n}{2} - F_c}{f_m} \times c$$

$$= 20 + \frac{15 - 11}{12} \times 10$$

$$= 23.33$$

$$\Rightarrow \text{Mode} = L + \frac{A_1}{A_1 + A_2} \times 10$$

$$= 20 + \frac{4}{4+7} \times 10$$

$$= 23.636$$

(ii) 1st quartiles, $Q_1 = L + \frac{\frac{1 \times n}{4} - F_c}{f_c} \times c$

$$= 10 + \frac{7.5 - 3}{8} \times 10$$

$$= 15.625$$

6th deciles, $D_6 = L + \frac{\frac{6 \times n}{10} - F_c}{f_c} \times c$

$$= 20 + \frac{18 - 11}{12} \times 10$$

$$= 25.83$$

78th percentiles, $P_{78} = L + \frac{\frac{78 \times n}{100} - F_p}{f_p} \times c$

$$= 30 + \frac{23.4 - 23}{5} \times 10$$

$$= 30.8$$

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